

GRAVITATIONAL DUALS FROM EQUATIONS OF STATE

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IBERIANSTRINGS 2025 GONG SHOW

BIZKAIA ARETOA-UPV/EHU, BILBAO, SPAIN



UNIVERSITAT DE
BARCELONA



Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA



EXCELENCIA
MARÍA
DE MAEZTU
2020 - 2024



MINISTERIO
DE CIENCIA, INNOVACIÓN
Y UNIVERSIDADES



AGENCIA
ESTATAL DE
INVESTIGACIÓN

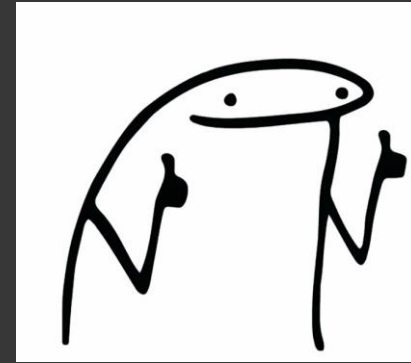
HOLOGRAPHY

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- Strongly interacting QFT $\leftrightarrow$ Classical Gravity in AdS

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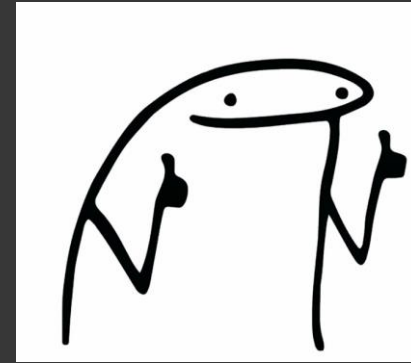
- Strongly interacting QFT $\leftrightarrow$ Classical Gravity in AdS
- Real time dynamics solving Einstein equations in 4+1 AdS spacetime:
 - Neutron Star mergers
 - Phase transitions in the early Universe



Good!

HOLOGRAPHY

- Strongly interacting QFT $\leftrightarrow$ Classical Gravity in AdS
- Real time dynamics solving Einstein equations in 4+1 AdS spacetime:
 - Neutron Star mergers
 - Phase transitions in the early Universe
- Very difficult to get the 4+1 AdS spacetime from the QFT properties

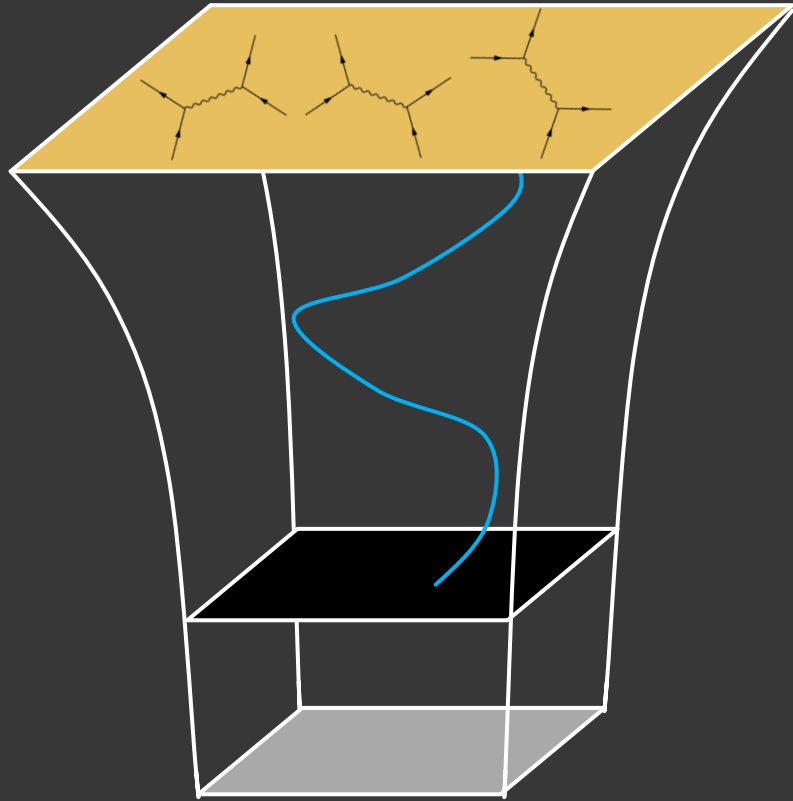


Good!

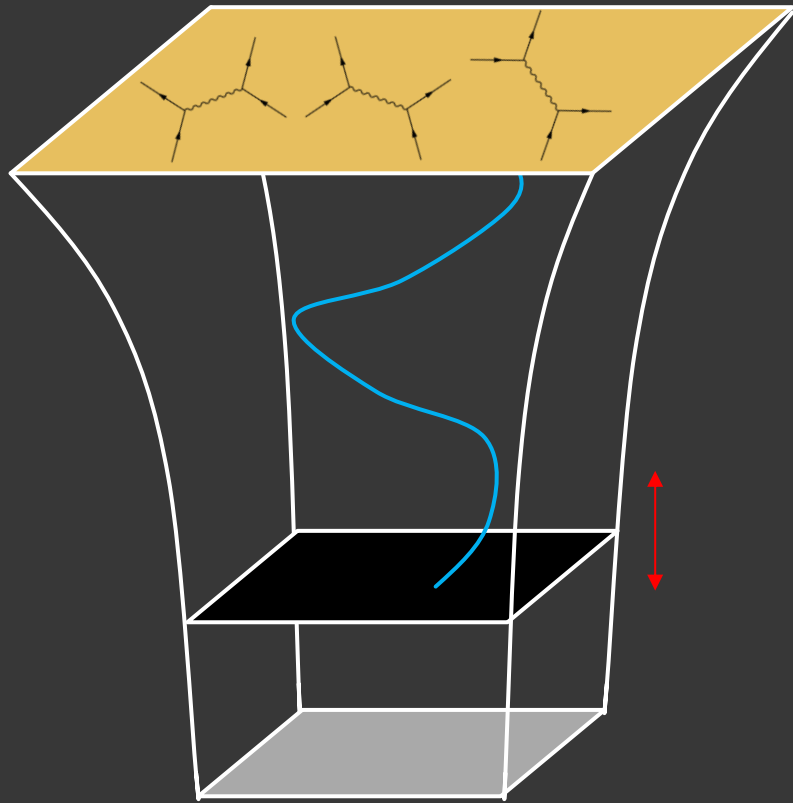


Bad
but...

HOLOGRAPHY IN A NUTSHELL

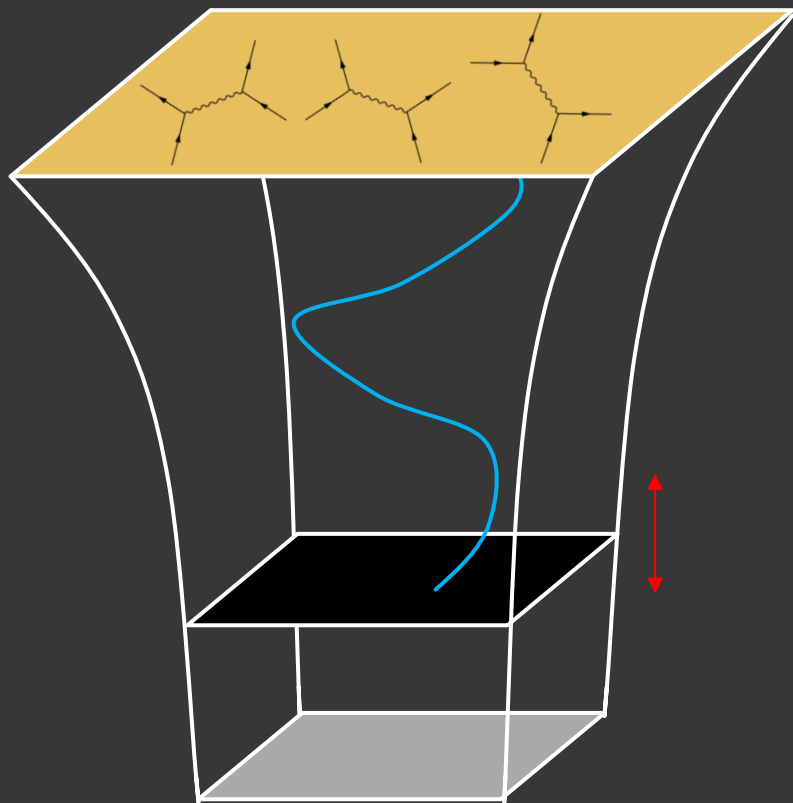


HOLOGRAPHY IN A NUTSHELL



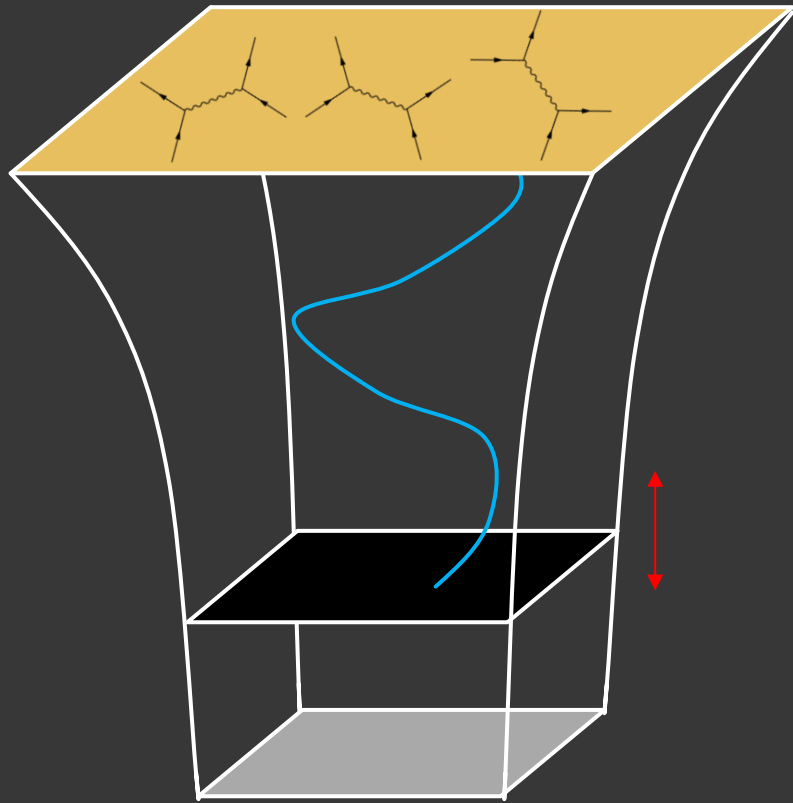
1. Construct different black brane solutions

HOLOGRAPHY IN A NUTSHELL



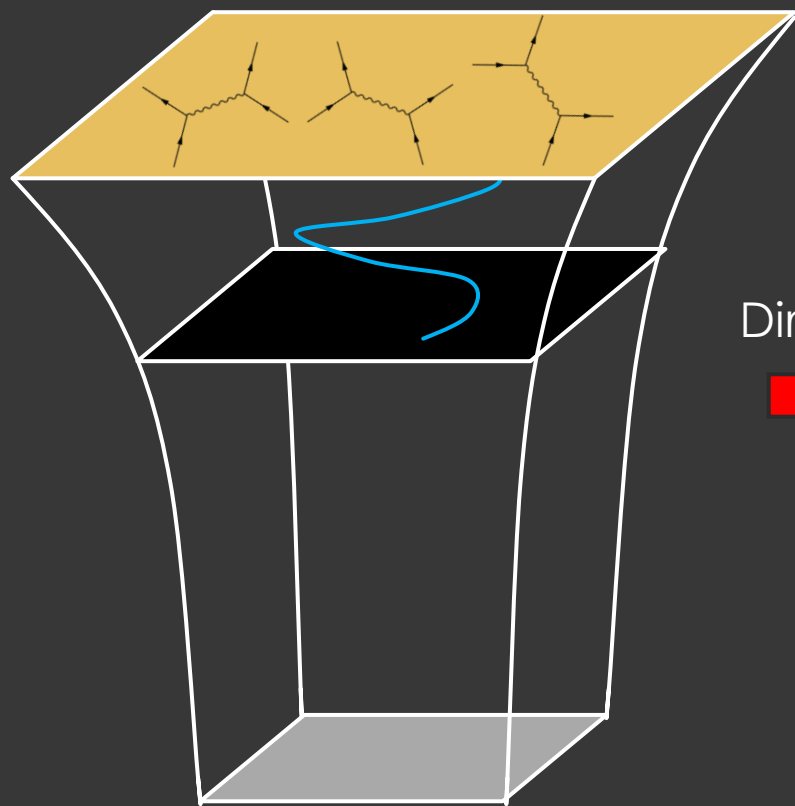
1. Construct different black brane solutions
2. Solve the EE+KG

HOLOGRAPHY IN A NUTSHELL

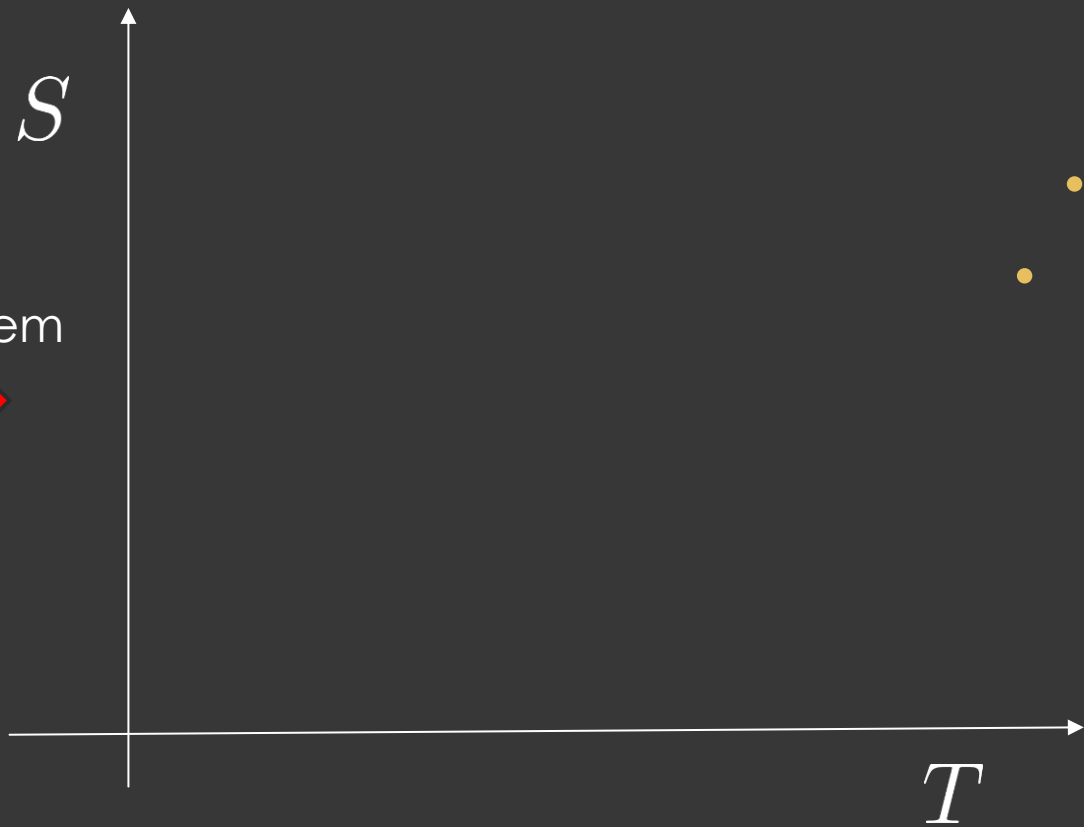


1. Construct different black brane solutions
2. Solve the EE+KG
3. Read off the thermodynamics as b.c.

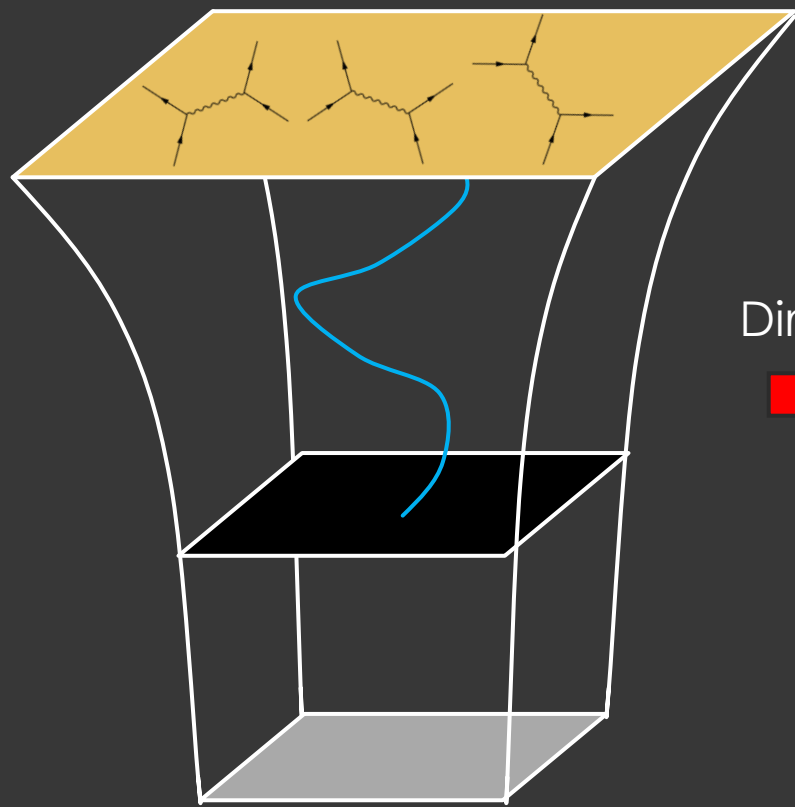
HOLOGRAPHY IN A NUTSHELL



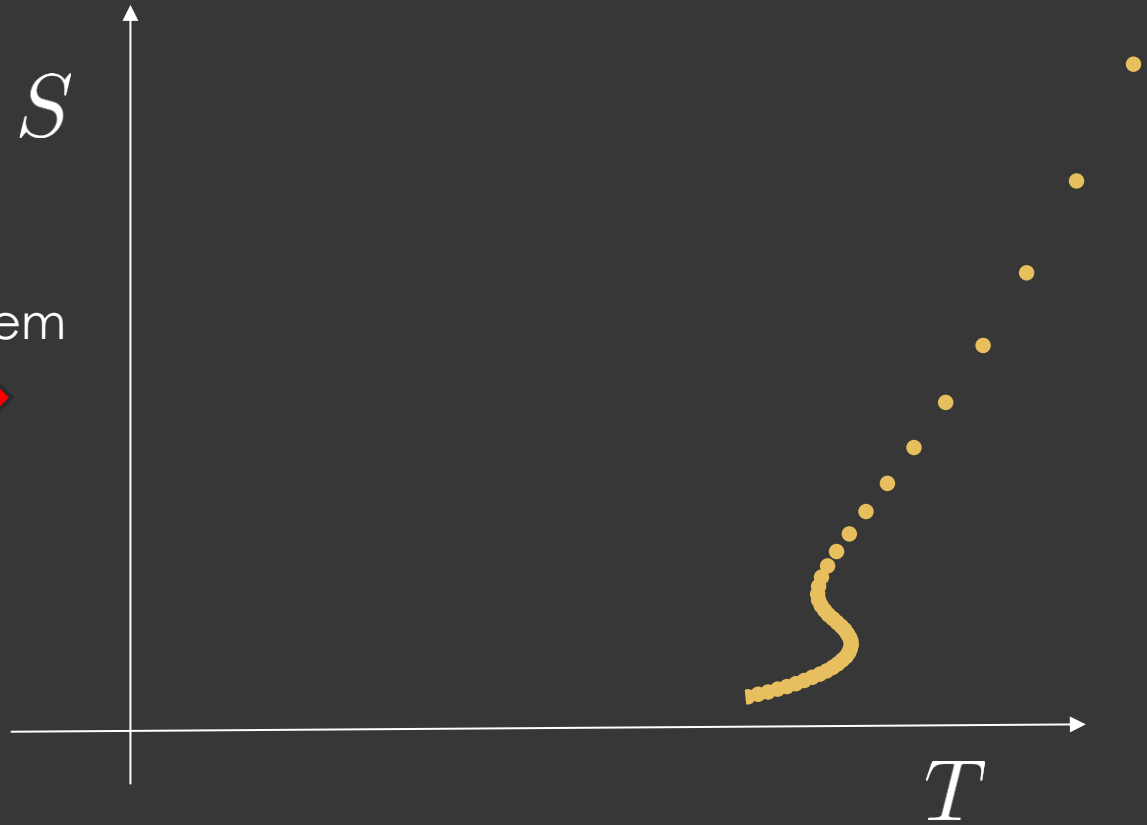
Direct problem



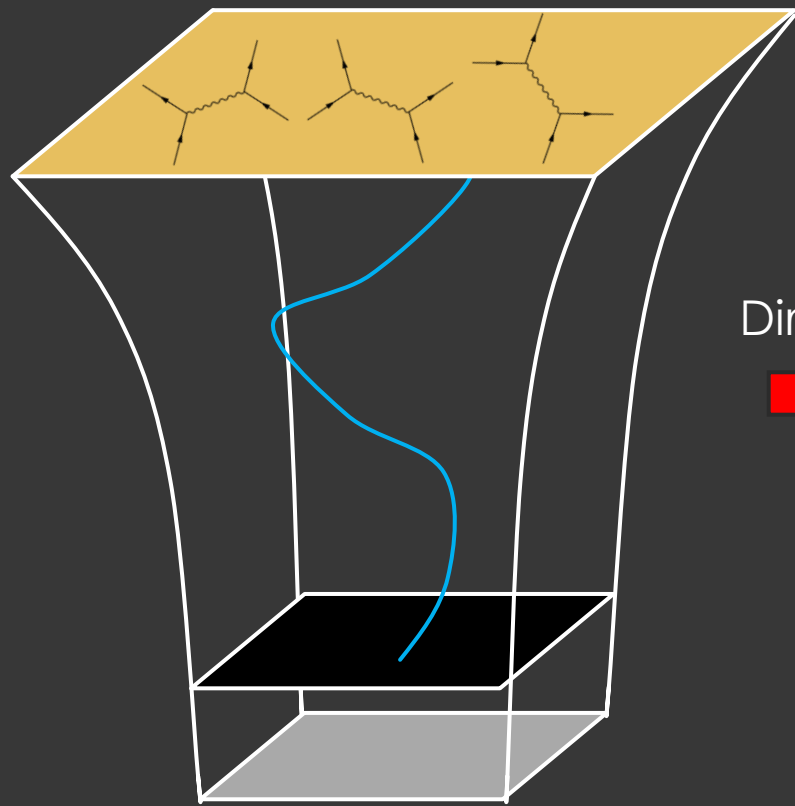
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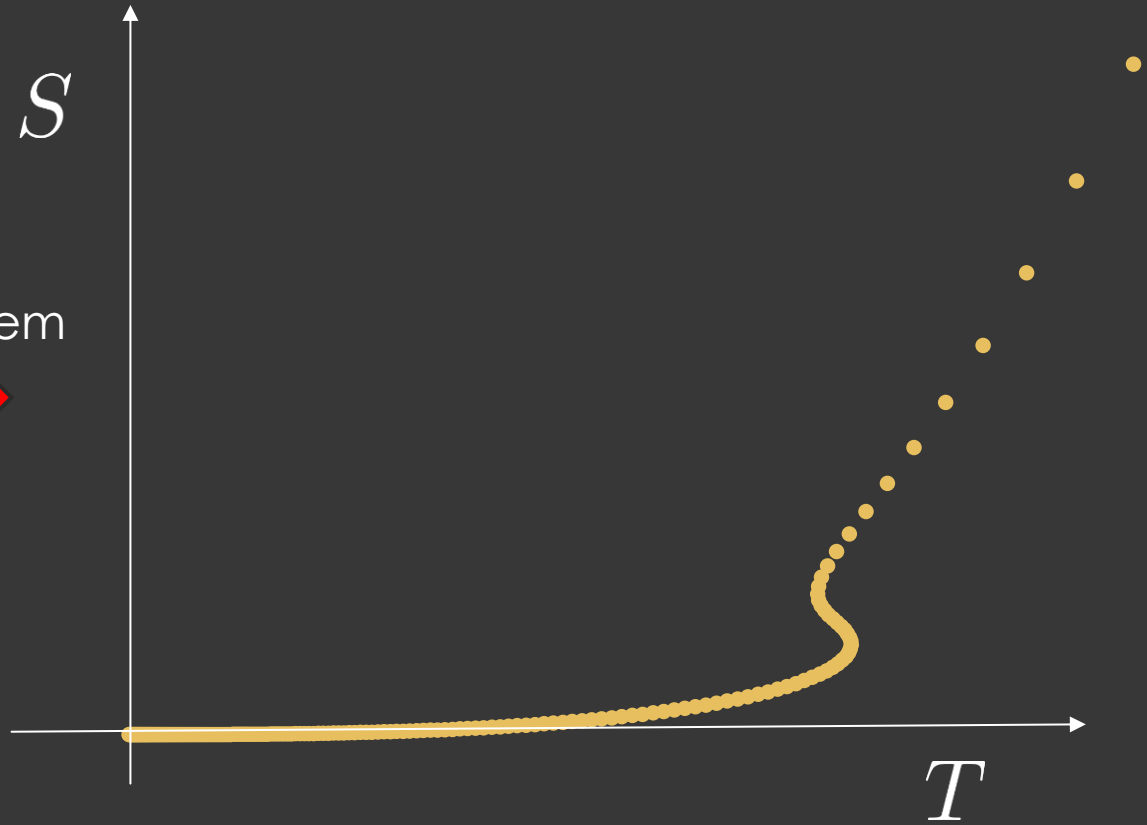
Direct problem



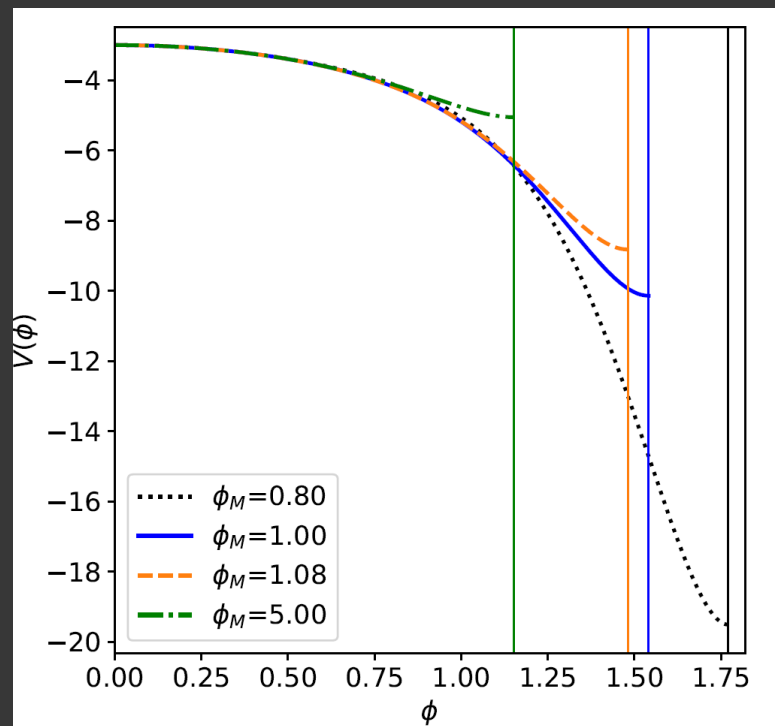
HOLOGRAPHY IN A NUTSHELL



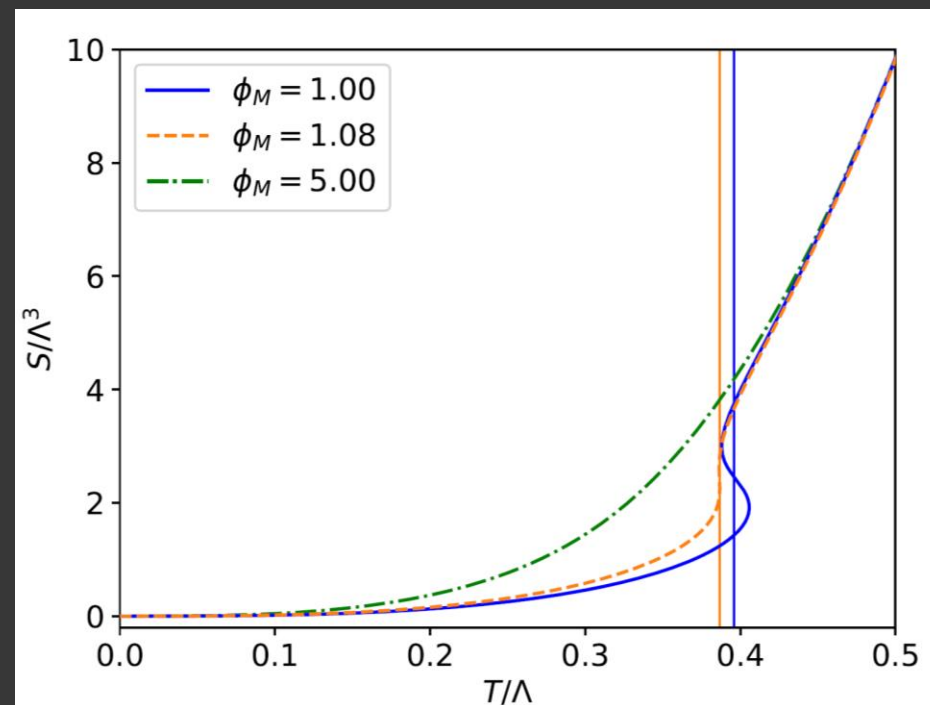
Direct problem



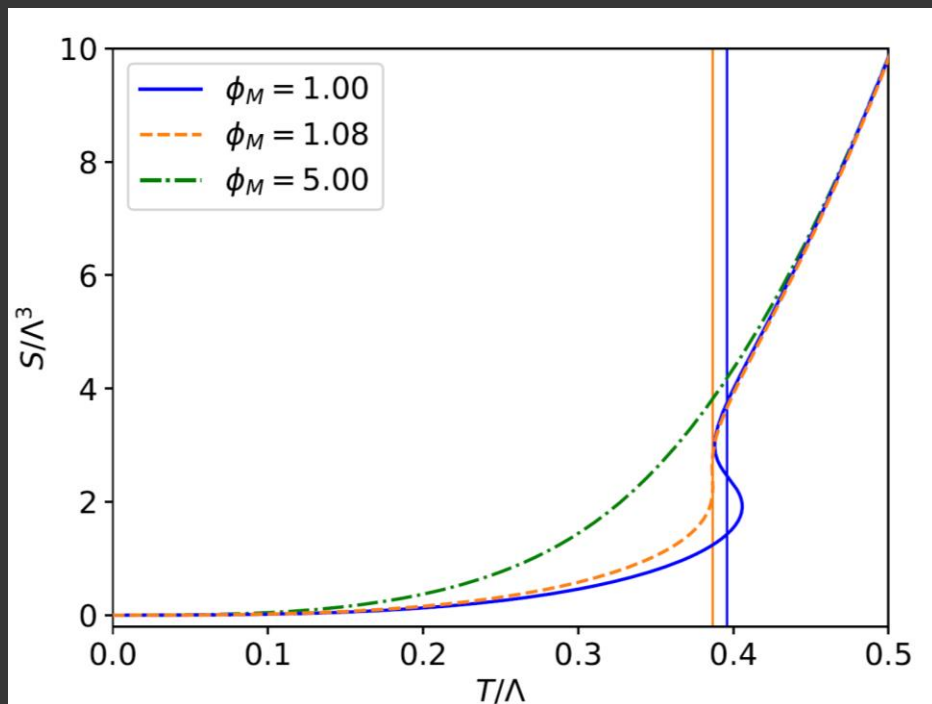
RESULTS



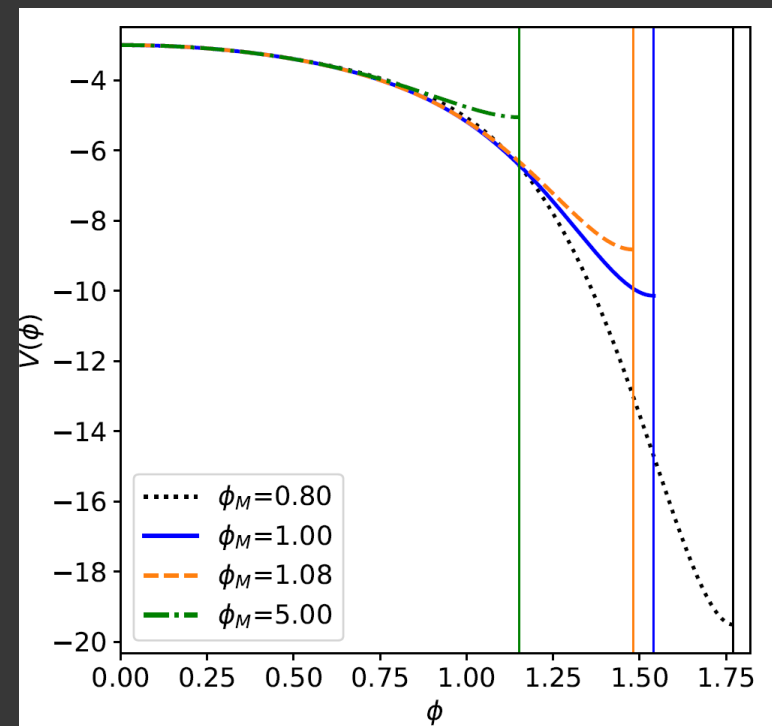
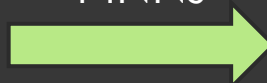
Direct problem



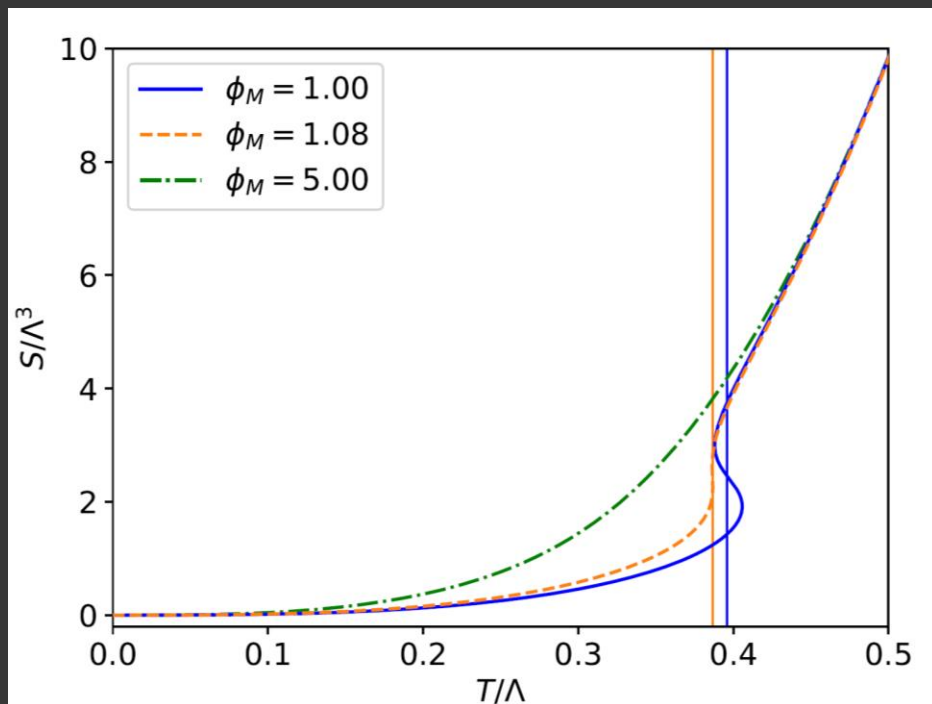
RESULTS



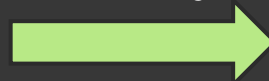
Inverse problem
PINNs



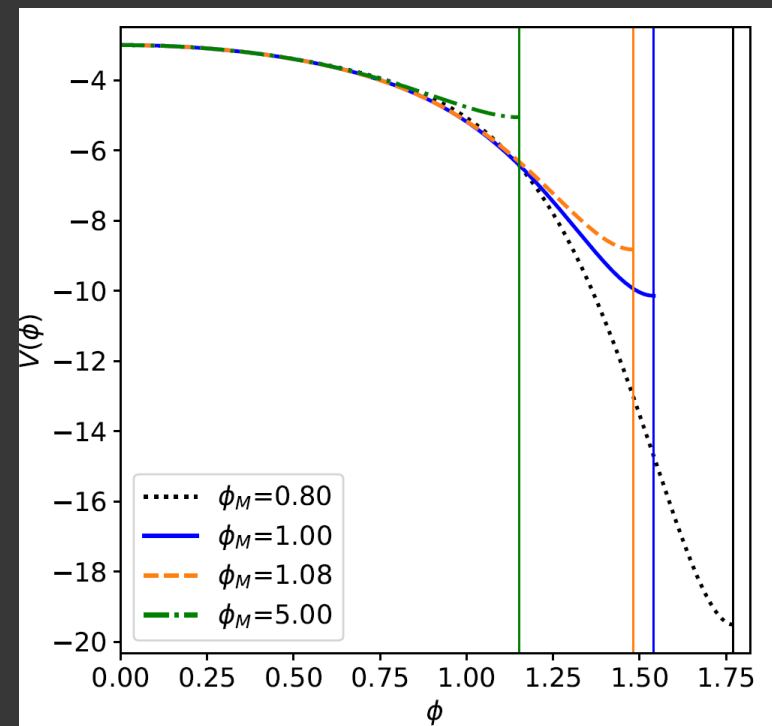
RESULTS



Inverse problem
PINNs



Direct problem
Compare $S(T)$

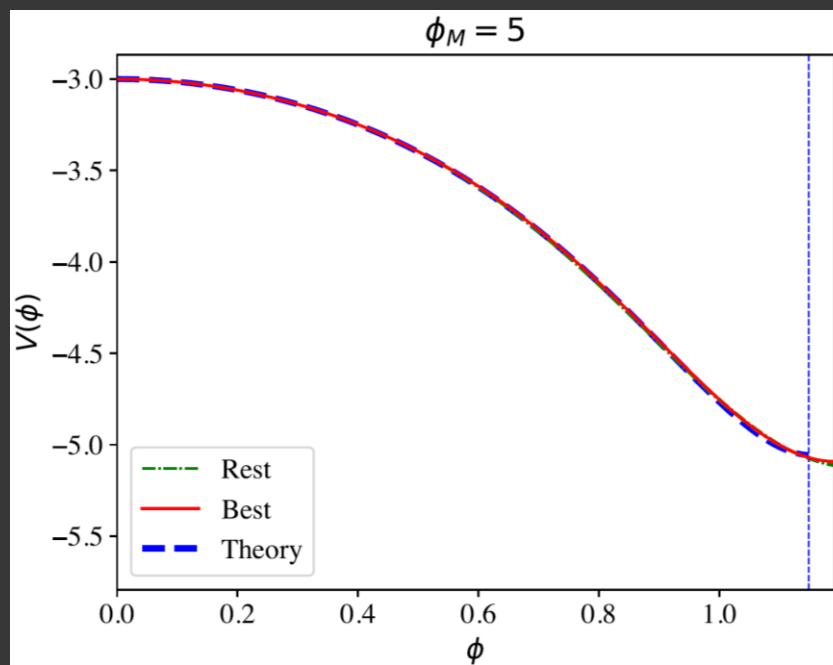


PHYSICS INFORMED NEURAL NETWORKS (PINNS)

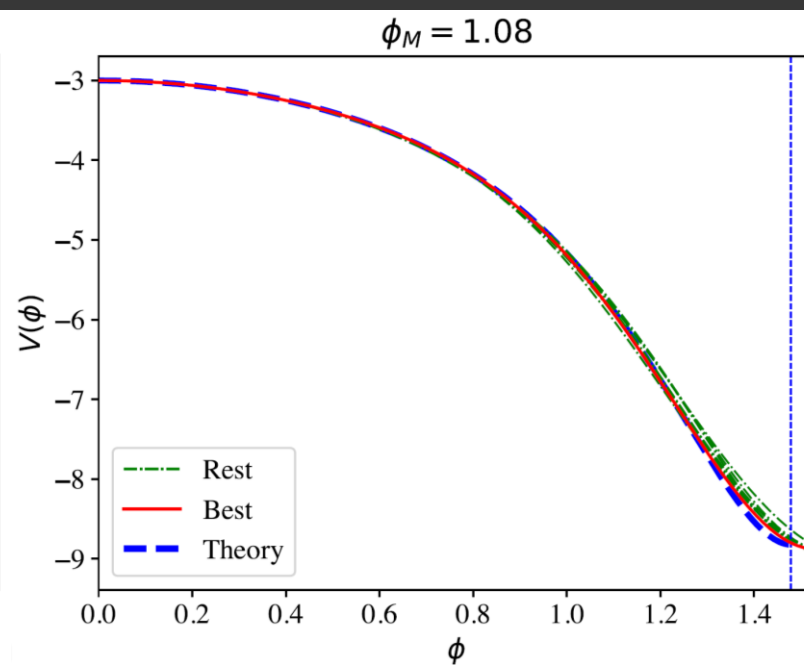
- Recover the potential + solve the DEs
- All the unknown functions are Neural Networks
- The potential is also a Neural Network, and thus a free function
- PINNs $\rightarrow$ minimize the residuals of the DEs

RESULTS

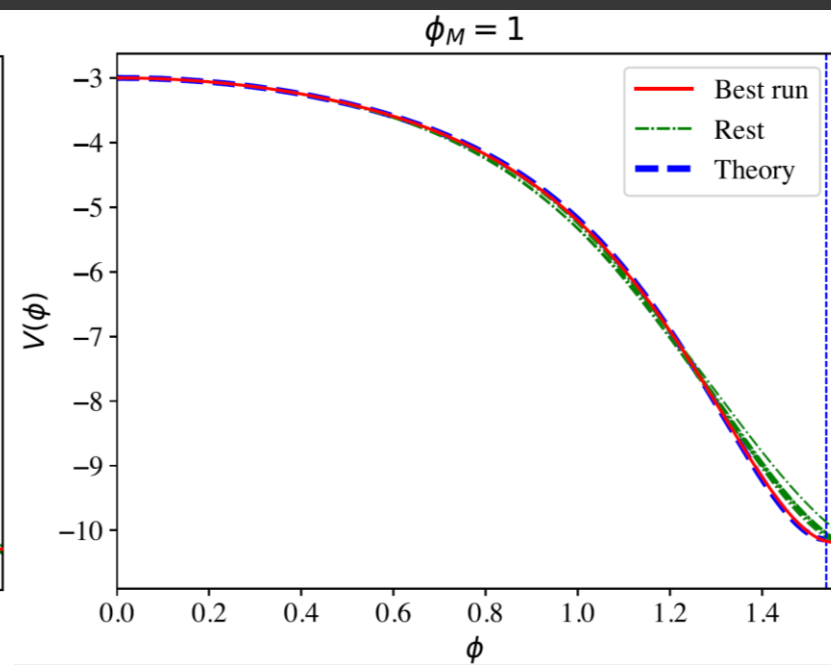
Crossover



Second Order

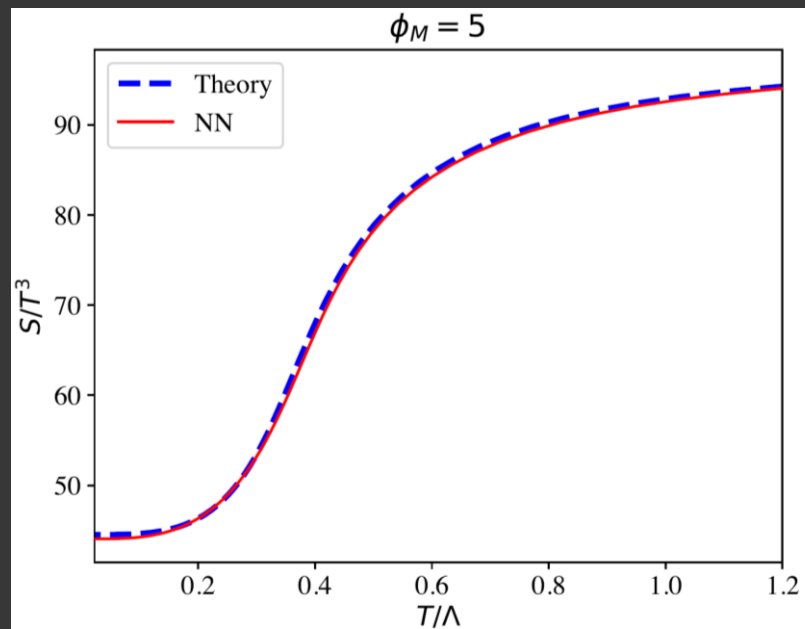


First Order

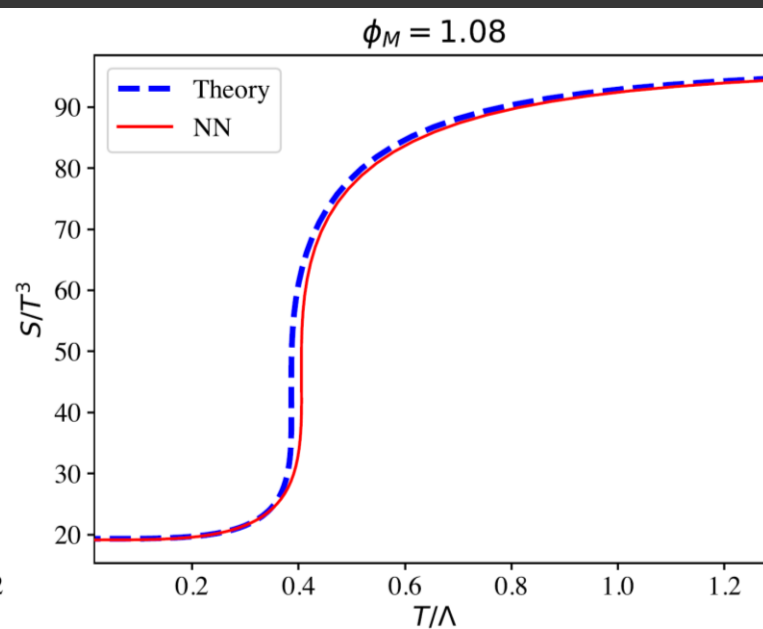


RESULTS

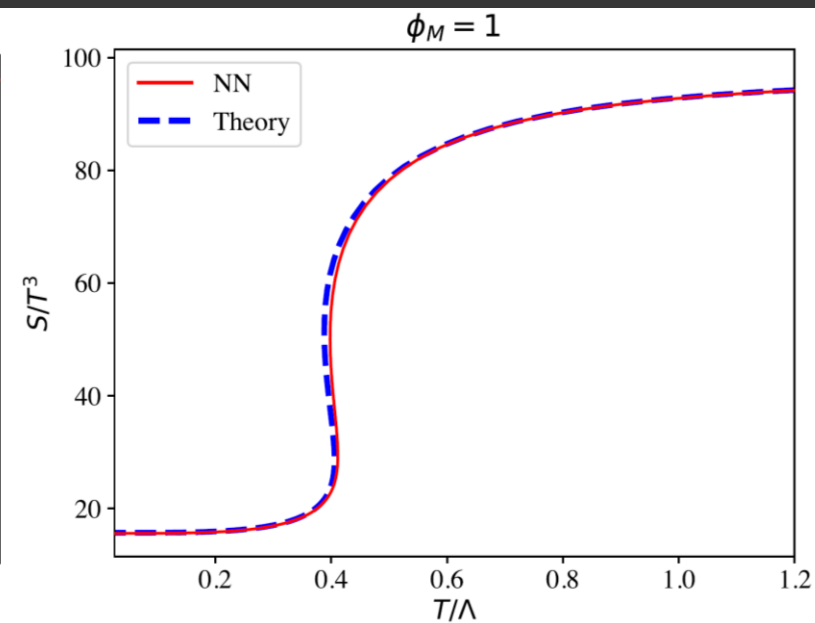
Crossover



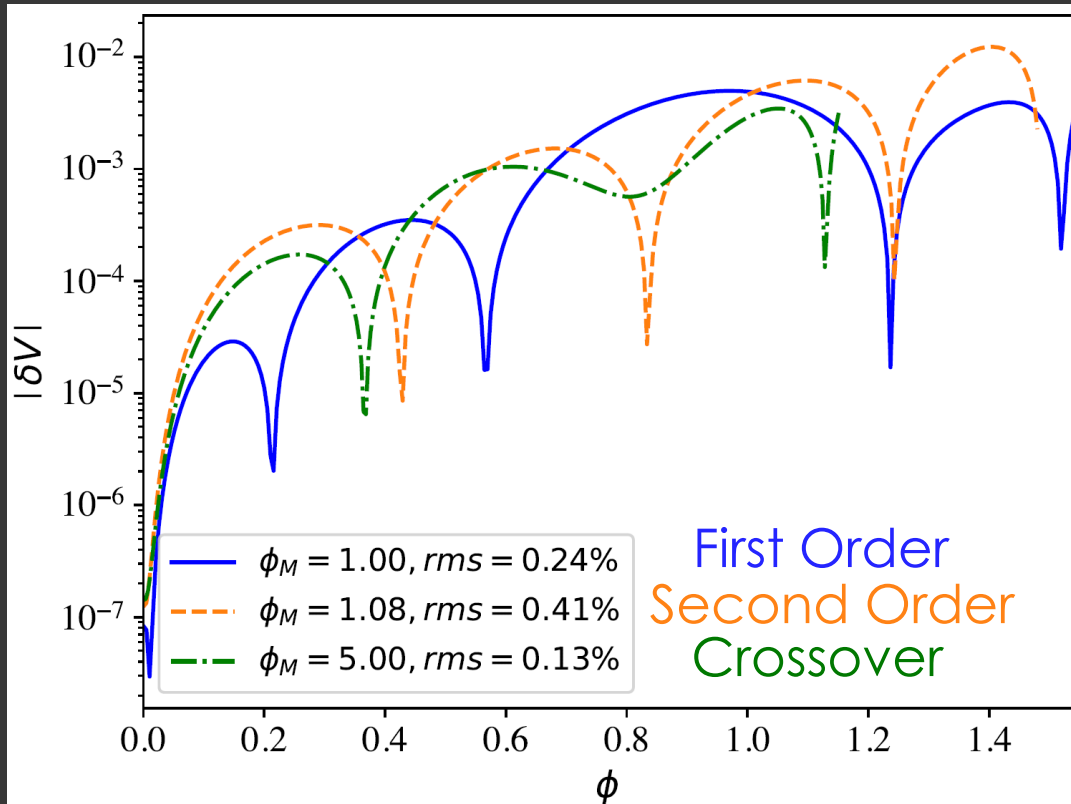
Second Order



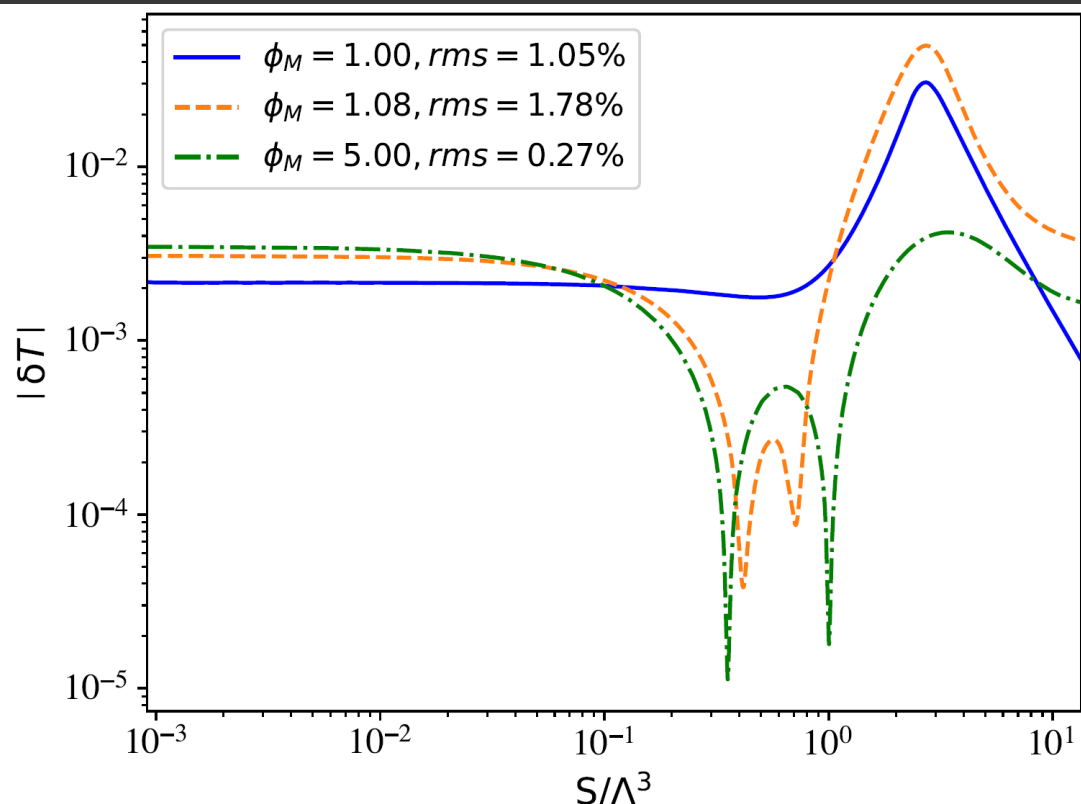
First Order



RESULTS



$$|\delta V(\phi)| = \left| \frac{V_{\text{theory}}(\phi) - V_{\text{PINN}}(\phi)}{V_{\text{theory}}(\phi)} \right|$$



$$|\delta T(S)| = \left| \frac{T_{\text{input}}(S) - T_{\text{PINN}}(S)}{T_{\text{input}}(S)} \right|$$

CONCLUSIONS AND FUTURE DIRECTIONS

- It is possible to recover the bulk gravitational theory using data from the boundary theory. This is a very non-trivial result in the field of Holography.
- Applicable to many situations: QCD equation of state, physics BSM from GW data, Neutron Star mergers...
- The setup is not good enough to solve the stiffest cases: false vacuum decay, naked singularity in the bulk...
- Method generalizable to other inversion problem in physics that involve highly non-linear differential equations, i.e. fluid dynamics.

THANK YOU!